Purnima Pandey

Bhopal, India | LinkedIn | Github | purnimap3002@gmail.com

Summary

Computer Science student with experience in Flutter, FastAPI, and MySQL, focused on building scalable full-stack applications and backend systems. Skilled in REST API development, authentication workflows, and database-driven application design. Interested in backend engineering and full-stack development roles.

Projects

Smart Hostel Attendance System | Github
Flutter, FastAPI, MySQL, OpenCV

- Developed a biometric hostel attendance system using OpenCV-based facial embeddings and cosine similarity for real-time identity verification
- Implemented multi-factor attendance validation using facial recognition, Wi-Fi SSID verification, and configurable time-window constraints
- Built secure RESTful APIs with FastAPI, JWT authentication, and role-based access control (RBAC) for students and administrators
- Designed modular backend architecture for attendance tracking, leave management, and access-controlled workflows
- Integrated Flutter frontend with backend services to enable seamless attendance monitoring and request handling

Appointment Booking System for Doctors | Github
Flutter, FastAPI, MySQL

- Built a full-stack doctor appointment platform using Flutter frontend and FastAPI backend architecture
- Designed slot-based scheduling workflows with conflict-free booking validation and dynamic appointment status updates
- Developed secure role-based access control (RBAC) and backend business rule validation for doctors and patients
- Built reusable Flutter widgets and integrated RESTful APIs to improve maintainability and user experience
- Deployed backend on Render with hosted database integration and documented REST APIs using Swagger

Skills

Languages- Python, Dart, Java, SQL
Frameworks & Technologies- Flutter, FastAPI, REST APIs, SQLAlchemy
Databases- MySQL, PostgreSQL
Tools- Git, GitHub, Swagger
Concepts- OOP, Database Design, Authentication & Authorization, RESTful Architecture, HTTP

Education

Lakshmi Narain College of Technology, Bhopal
August 2023 - Present | Bachelor of Technology
Computer Science and Engineering | CGPA- 8.76
Vidya Vihar Residential School, Purnea
April 2021 - March 2023| Higher Secondary Certificate (CBSE)

Achievements

Runner-up | Statathon-2025
Collaborated in a 4-member team to build an ML- powered license plate recognition system using OpenCV and TensorFlow. Competed against 50+ participants and won for innovation.
Cognizant Digital Nurture- 5.0
Shortlisted for the DeepSkilling phase of Cognizant Digital Nurture 5.0 Java FSE programme, progressing through the initial screening rounds.

Certifications

Google Flutter-Advanced | Infosys Springboard
Database Management Systems | Infosys Springboard
Data Structures and Algorithms using Java | Infosys Springboard